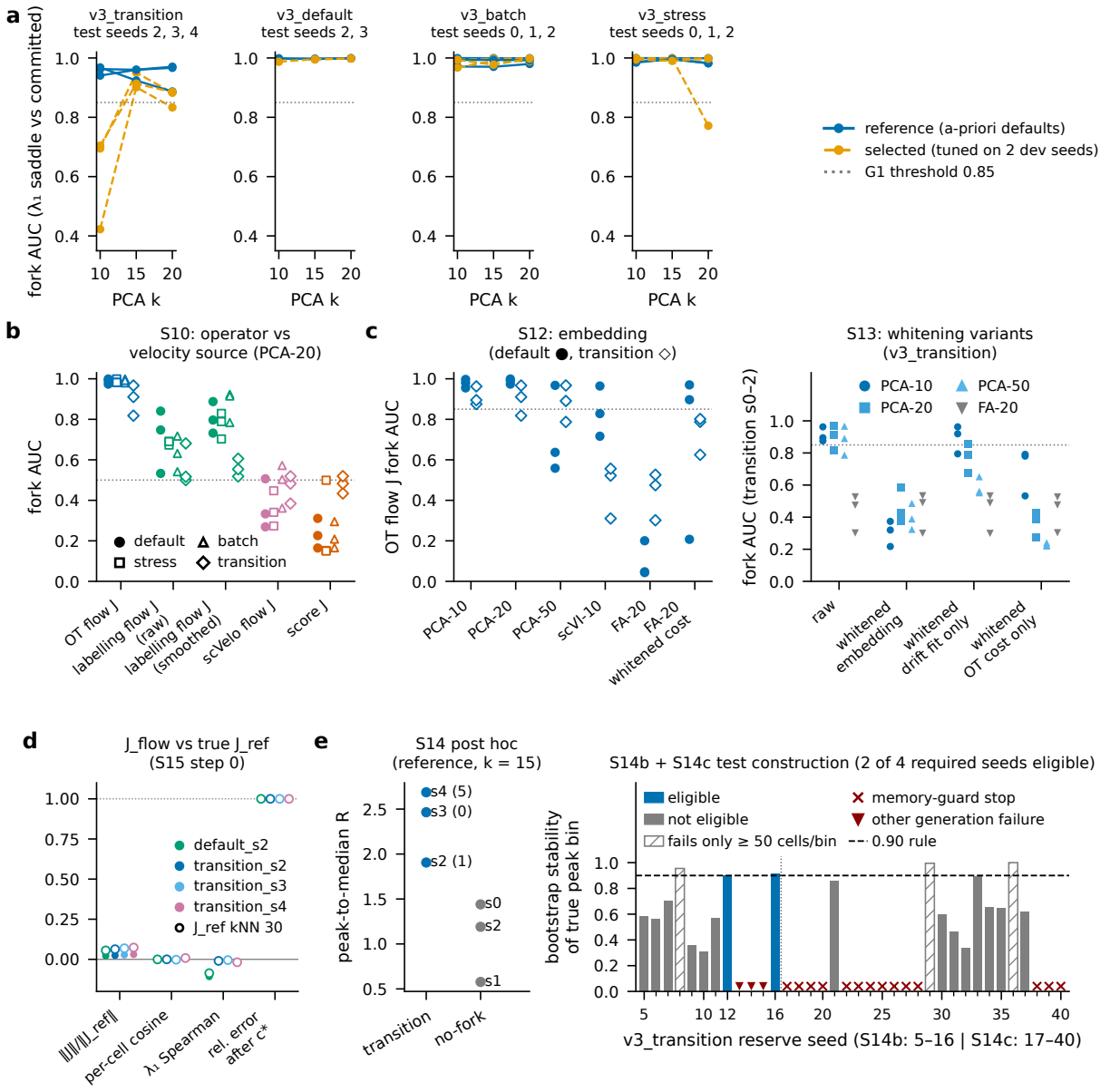

a
v3_transition
test seeds 2, 3, 4
v3_default
test seeds 2, 3
v3_batch
test seeds 0, 1, 2
v3_stress
test seeds 0, 1, 2
fork AUC (λ1 saddle vs committed)
PCA k
reference (a-priori defaults)
selected (tuned on 2 dev seeds)
G1 threshold 0.85
b
S10: operator vs
velocity source (PCA-20)
fork AUC
default
batch
stress
transition
OT flow J
labelling flow J (raw)
labelling flow J (smoothed)
scVelo flow J
score J
c
S12: embedding
(default ●, transition ◇)
OT flow J fork AUC
PCA-10
PCA-20
PCA-50
scVI-10
FA-20
FA-20 whitened cost
S13: whitening variants
(v3_transition)
fork AUC (transition s0–2)
PCA-10
PCA-20
PCA-50
FA-20
raw
whitened embedding
whitened drift fit only
whitened OT cost only
d
J_flow vs true J_ref
(S15 step 0)
default_s2
transition_s2
transition_s3
transition_s4
J_ref kNN 30
‖J‖/‖J_ref‖
per-cell cosine
λ1 Spearman
rel. error after c*
e
S14 post hoc
(reference, k = 15)
peak-to-median R
s4 (5)
s3 (0)
s2 (1)
s0
s2
s1
transition
no-fork
S14b + S14c test construction (2 of 4 required seeds eligible)
eligible
not eligible
fails only ≥ 50 cells/bin
memory-guard stop
other generation failure
0.90 rule
bootstrap stability of true peak bin
v3_transition reserve seed (S14b: 5–16 | S14c: 17–40)